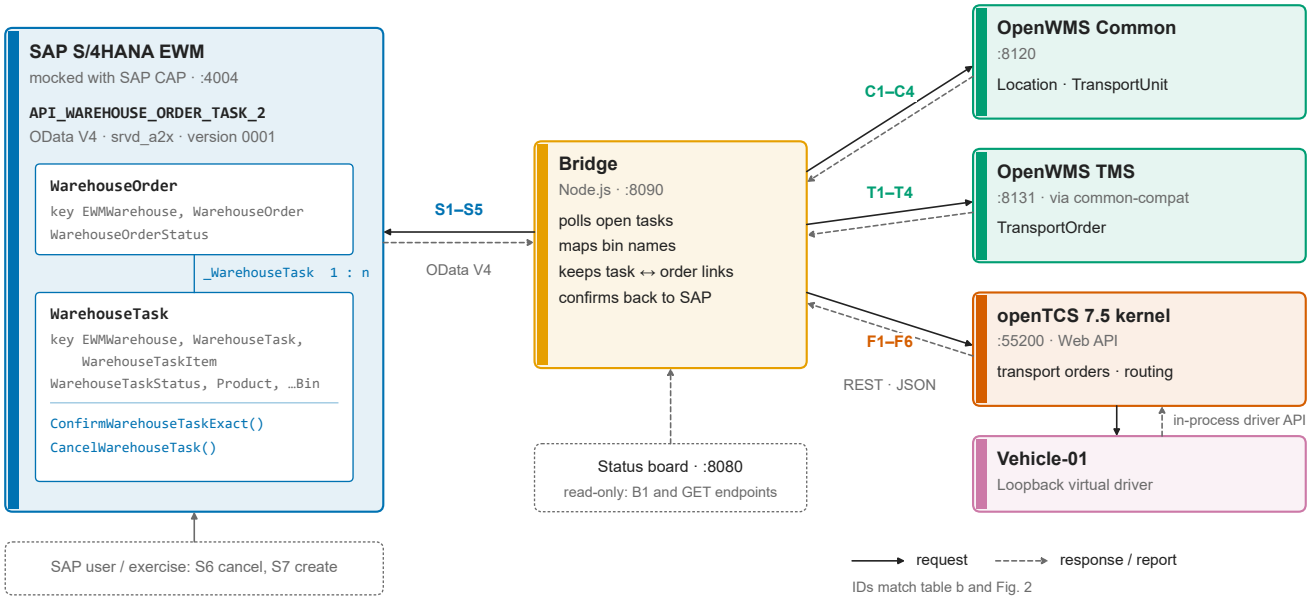
a



b

ID	Method	Path	Purpose	Phase
SAP EWM · {EWM} = /sap/opu/odata4/sap/api_warehouse_order_task_2/srvd_a2x/sap/warehouseorder/0001				
(…) = (EWMWarehouse='E001',WarehouseTask='1000000001',WarehouseTaskItem='0'); writes need X-CSRF-Token and If-Match				
S1	GET	{EWM}/$metadata	readiness, metadata	①
S2	GET	{EWM}/WarehouseTask?$filter=WarehouseTaskStatus eq ''	poll open warehouse tasks	②
S3	GET	{EWM}/WarehouseTask(…)	read status and ETag	⑤ ⑥
S4	HEAD	{EWM}/ X-CSRF-Token: Fetch	fetch CSRF token	⑥
S5	POST	{EWM}/WarehouseTask(…)/SAP__self.ConfirmWarehouseTaskExact	confirm warehouse task	⑥
S6	POST	{EWM}/WarehouseTask(…)/SAP__self.CancelWarehouseTask	cancel warehouse task	SAP side
S7	POST	{EWM}/WarehouseTask	create warehouse task	SAP side
OpenWMS Common · http://openwms-common:8120				
C1	GET	/actuator/info	readiness	①
C2	GET POST	/v1/locations?locationId=…	find / create location	①
C3	GET POST	/v1/transport-units?bk=…[&actualLocation=…&tut=PALLET]	find / create pallet	①
C4	PATCH	/v1/transport-units?bk=…&newLocation=…	move pallet	③ ⑥
OpenWMS TMS · http://openwms-tms:8131				
T1	GET	/actuator/info	readiness	①
T2	POST	/v1/transport-orders	create order, 201 + Location	③
T3	GET	/v1/transport-orders/{pKey}	read order state	③ ⑥
T4	POST	/v1/transport-orders/{pKey}?state=STARTED|FINISHED|ONFAILURE|CANCELED	change order state	③ ⑥
openTCS · http://opentcs:55200				
F1	GET	/v1/kernel/version	readiness	①
F2	GET PUT	/v1/plantModel	read / upload plant model	①
F3	PUT	/v1/vehicles/Vehicle-01/commAdapter/{attachment,enabled}, …/integrationLevel	attach Loopback, allow orders	①
F4	POST	/v1/transportOrders/SAP-{WarehouseTask}	create order: load → unload	④
F5	GET	/v1/transportOrders/SAP-{WarehouseTask}	read order state	⑤
F6	POST	/v1/transportOrders/SAP-{WarehouseTask}/withdrawal?immediate=false	withdraw order	cancel
Bridge · http://bridge:8090				
B1	GET	/status /health	progress, links, journal	observe

Phases: ① init ② query ③ WMS order ④ fleet order ⑤ monitor ⑥ confirm (see Fig. 2). Vehicle and kernel talk through an in-process driver API, not HTTP.